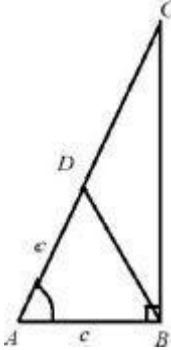
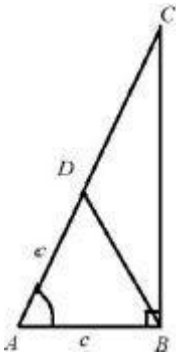


Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses area of ADB = area of CDB or area of ADB = $\frac{1}{2}$ area of ABC Possibly by use of " $\frac{1}{2}ab \sin C$ " twice	AO3.1a	M1	 Area of ADB = Area of CDB $CD = AD = c$ $AC = 2c$ $\cos A = \frac{c}{2c} = \frac{1}{2}$ so $A = 60^\circ$ $\tan A = \sqrt{3}$ $\sin A = \frac{\sqrt{3}}{2}$ $2\sin A = \sqrt{3} = \tan A$
	Deduces that $AC = 2 \times AD = 2 \times AB$ or equivalent	AO1.1b	A1	
	Uses trigonometry involving sin and tan based on triangle with $AC = 2 \times AB$	AO1.1a	M1	
	Obtains correct conclusion (AG) Sets out a well-constructed mathematical argument. Use of 60° or equivalent must be justified	AO2.1	R1	
(b) (i)	Uses $\tan A = \frac{\sin A}{\cos A}$ and multiplies Or Uses sketch of two graphs to show two intersections	AO1.1a	M1	$\frac{\sin A}{\cos A} = 2 \sin A$ $\sin A = 2 \sin A \cos A$ $\sin A(1 - 2 \cos A) = 0$ $\sin A = 0$ or $\cos A = \frac{1}{2}$ $A = 0^\circ$ and $A = 60^\circ$
	Solves the equation to give $A = 0^\circ$ and 60° Or interprets intersections of graphs of the correct shape between 0° and 90° to be the solutions Special case 0° and 60° stated but not justified award B1. Stated and verified award B2	AO1.1b	A1	
(ii)	Selects $A = 60^\circ$. (Can be earned with no other	AO3.2a	B1	We need $A = 60^\circ$

working shown)			
Alternative marking Instructions	AO	Marks	Typical Solution
(a) Obtains area formula for ABD using $\sin A$	AO3.1a	M1	 Area of $ADB = \frac{1}{2}c^2 \sin A$
Obtains expression for BC using $\tan A$	AO1.1a	M1	$\frac{BC}{c} = \tan A$ $BC = c \tan A$
Obtains correct expression for area of ABC	AO1.1b	A1	Area of $ABC = \frac{1}{2}c^2 \tan A$
Simplifies to correct conclusion (AG) Sets out a well-constructed mathematical argument.	AO2.1	R1	$\frac{1}{2}c^2 \tan A = 2 \times \frac{1}{2}c^2 \sin A$ $\tan A = 2 \sin A$
(b) (i) Uses $\tan A = \frac{\sin A}{\cos A}$ and multiplies Or Uses sketch of two graphs to show two intersections	AO1.1a	M1	$\frac{\sin A}{\cos A} = 2 \sin A$ $\sin A = 2 \sin A \cos A$
Solves the equation to give $A = 0^\circ$ and 60° Or interprets intersections of graphs of the correct shape between 0° and 90° to be the solutions Special case 0° and 60° stated but not justified award B1. Stated and verified award B2	AO1.1b	A1	$\sin A(1 - 2 \cos A) = 0$ $\sin A = 0 \text{ or } \cos A = \frac{1}{2}$ $A = 0^\circ \text{ and } A = 60^\circ$
(ii) Selects $A = 60^\circ$. (Can be earned with no other working shown)	AO3.2a	B1	We need $A = 60^\circ$
Total 7 marks			

Q2.

Marking Instructions	AO	Marks	Typical Solution
Solves to find $\tan 2\theta$ (allow M1 even if only $+\sqrt{3}$ given PI by any correct final value of θ)	AO1.1a	M1	$\tan 2\theta = \pm\sqrt{3}$ $\tan 2\theta = \sqrt{3} \rightarrow 2\theta = 60^\circ, 240^\circ, 420^\circ, 600^\circ$ $\tan 2\theta = -\sqrt{3} \rightarrow 2\theta = 120^\circ, 300^\circ, 480^\circ, 660^\circ$ $\theta = 30^\circ, 60^\circ, 120^\circ, 150^\circ, 210^\circ, 240^\circ, 300^\circ, 330^\circ$
Obtains at least 3 correct final values for θ ignore extra incorrect terms or terms outside range	AO1.1b	A1	
Obtains at least 3 correct final values of θ from $\tan 2\theta = \sqrt{3}$ and at least 3 correct values from $\tan 2\theta = -\sqrt{3}$ ignore extra incorrect terms or terms outside range	AO1.1a	M1	
Obtains complete set of exactly 8 correct values for θ	AO1.1b	A1	
			Total 4 marks

Q3.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	4
Total 1 mark			

Q4.

Marking Instructions	AO	Marks	Typical Solution
(a) Uses $A = \frac{1}{2}ab\sin C$ for triangle OAC or OAB PI by equation	AO1.2	B1	$\frac{1}{2}r \times \frac{r}{2} \sin \theta = \frac{1}{4} \left(\frac{1}{2} r^2 \theta \right)$ $\Rightarrow \frac{r^4}{4} \sin \theta = \frac{1}{8} r^2 \theta$ $\Rightarrow 2r^2 \sin \theta = r^2 \theta$ $\Rightarrow 2 \sin \theta = \theta$ AG
Forms an equation relating the area of OAC and ABC in the form $Ar^2 \sin \theta = Br^2 \theta$	AO3.1a	M1	

	Obtains fully correct equation ACF	AO1.1b	A1	
	Simplifies to obtain required equation, only award if all working correct with rigorous argument.	AO2.1	R1	
(b)	Rearranges to the form $f(\theta) = 0$ PI by correct θ_2 or θ_3	AO1.1a	M1	$f(\theta) = \theta - 2 \sin \theta = 0$ $\theta_{n+1} = \theta_n - \frac{\theta_n - 2 \sin \theta_n}{1 - 2 \cos \theta_n}$
	Differentiates their $f(\theta)$ or uses calculator PI correct θ_2 or θ_3	AO1.1b	A1	$\theta_2 = 2.094395\dots$ $\theta_3 = 1.913222\dots$ $\theta_3 = 1.91322$ (5 d.p.)
	Obtains correct θ_3	AO1.1b	A1	
(c)	Obtains percentage error for θ_3 AWRT 0.94%	AO3.2b	B1	0.935%
Total 8 marks				

Q5.

Marking Instructions	AO	Marks	Typical Solution
Translates given information into an equation by using the formula for the area of triangle or parallelogram to form a correct equation	AO3.1a	M1	$AB \times AD \times \sin \alpha = 24$ hence $6 \times 4.5 \times \sin \alpha = 24$
Rearranges 'their' equation to obtain a correct value of $\sin \alpha$	AO1.1b	A1F	$\sin \alpha = \frac{24}{27} = \frac{8}{9}$
Uses 'their' $\sin \alpha$ value to identify an appropriate right-angled triangle or uses identities and deduces exact ratio of $\tan \alpha$ – positive or negative Condone only positive ratio seen	AO2.2a	M1	Sides of right angled triangle are 8, 9 and $\sqrt{17}$ $\tan \alpha = \pm \frac{8}{\sqrt{17}}$ Hence
Relates back to mathematical context of	AO3.2a	A1F	α is one of the largest angles and must be obtuse

problem and hence chooses negative ratio – accept any equivalent exact form FT ‘their’ tan values for obtuse α			hence tangent is negative $\tan \alpha = -\frac{8}{\sqrt{17}} = -\frac{8\sqrt{17}}{17}$
Total 4 marks			

Q6.

Marking Instructions	AO	Marks	Typical Solution
Divides or multiplies by $\cos \theta$	AO3.1a	M1	$\frac{\sin \theta \tan \theta}{\cos \theta} + 2 \frac{\sin \theta}{\cos \theta} = 3$
Obtains correct quadratic	AO1.1b	A1	$\tan^2 \theta + 2 \tan \theta - 3 = 0$
Applies a correct method to solve ‘their’ quadratic PI	AO1.1a	M1	$(\tan \theta + 3)(\tan \theta - 1) = 0$ $\tan \theta = 1$ or -3
Finds two correct values of $\tan \theta$ from ‘their’ quadratic	AO1.1b	A1F	$\theta = 45^\circ$ or 108° ALT
Obtains two correct answers CAO	AO1.1b	A1	$\sin \theta \tan \theta \cos \theta + 2 \sin \theta \cos \theta = 3 \cos^2 \theta$ $\sin^2 \theta + 2 \sin \theta \cos \theta - 3 \cos^2 \theta = 0$ $(\sin \theta + 3 \cos \theta)(\sin \theta - \cos \theta) = 0$ $\tan \theta = 1$ or -3 $\theta = 45^\circ$ or 108°
Total 5 marks			

Q7.

Marking Instructions	AO	Marks	Typical Solution
(a) Recalls $A = \frac{1}{2} r^2 \theta$ or $l = r\theta$ PI by use in equation	AO1.2	B1	Area of sector gives $\frac{1}{2} r^2 \theta = 9$, $\theta = \frac{18}{r^2}$
Constructs two equations at least one correct	AO1.1a	M1	Perimeter of sector gives $2r + r\theta = 15$

	Eliminates θ FT incorrect equations	AO1.1a	M1	$2r + \frac{18}{r} = 15$
	Constructs a rigorous mathematical argument to show required result, clearly constructing two correct simultaneous equations and eliminating θ AG	AO2.1	R1	$2r^2 + 18 = 15r$ $2r^2 - 15r + 18 = 0$ (AG)
(b)	Solves a quadratic equation and finds two values of θ	AO3.1a	M1	$r = \frac{3}{2}, r = 6$
	Finds two correct values of r	AO1.1b	B1	$r = 6 \Rightarrow \theta = \frac{1}{2}$ $r = \frac{3}{2} \Rightarrow \theta = 8$
	Finds both values of θ	AO1.1b	A1	$8 > 2\pi \therefore \theta \neq 8$
	Gives a valid reason for rejecting one of 'their' values	AO2.4	R1	so only one possible value of θ
Total 8 marks				

Q8.

Marking Instructions	AO	Marks	Typical Solution
Recalls a correct trig identity, which could lead to a correct answer	AO1.2	B1	(LHS $\equiv$) $\cot^2 \theta - \cos^2 \theta$
Performs some correct algebraic manipulation and uses second identity to commence proof (at least two lines of argument)	AO2.1	R1	$\frac{\cos^2 \theta}{\sin^2 \theta} - \cos^2 \theta$ $\equiv \cos^2 \theta \left(\frac{1}{\sin^2 \theta} - 1 \right)$ $\equiv \cos^2 \theta (\operatorname{cosec}^2 \theta - 1)$ $\equiv \cos^2 \theta \cot^2 \theta$ ($\equiv$ RHS)
Concludes a rigorous mathematical argument to prove given identity AG Must start with one side and through clear logical steps arrive at the other side. In order to be sufficiently clear, each line should be a single step, unless clear further	AO2.1	R1	AG

explanation is given.			
Total 3 marks			

Q9.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Makes clear attempt to use the cosine rule	AO3.1a	M1	$6^2 = 3^2 + 5^2 - 2 \times 3 \times 5 \cos \theta$
	Uses trig identity with 'their' $\cos \theta$	AO1.1a	M1	$\cos \theta = \frac{3^2 + 5^2 - 6^2}{30} = -\frac{1}{15}$
	Constructs rigorous argument leading to correct result AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	$\therefore \sin \theta = \sqrt{1 - \left(-\frac{1}{15}\right)^2}$ $\sin \theta = \frac{4\sqrt{14}}{15} \quad \textbf{(AG)}$
(b)	Writes down correct angle	AO2.2a	B1	1.64
(c)	Uses 'their' angle in $\frac{1}{2}r^2\theta$	AO1.1a	M1	$A = \frac{1}{2} \times 5^2 \times 1.64$
	Correct area FT use of incorrect obtuse angle provided both M1 marks awarded in part (a) and M1 awarded in (c)	AO1.1b	A1F	$= 20.5 \text{ m}^2$
				Total 6 marks

Q10.

- (a) Arc = $r\theta$ ($= 1.25r$)
Within (a), $r\theta$ or 15 used for the arc length PI

M1

$$P = r + r + r\theta = 39$$

Use of $r + r + r\theta$ for the perimeter.

m0 if no indication that '15' comes from $r\theta$

m1

$$3.25r = 39 \quad r = \frac{39}{3.25} = 12$$

CSO AG

A1

3

(b) {Area of sector = } $\frac{1}{2}r^2\theta$

Within (b), $\frac{1}{2}r^2\theta$ stated or used for the sector area.

M1

$$= \frac{1}{2} \times 12^2 \times 1.25 = 90 \text{ (cm}^2\text{)}$$

NMS: 90 scores 2 marks

A1

2

[5]

Q11.

(a) $\frac{1}{2} \times 5 \times 6 \sin C = 12.5$

(Area =) $\frac{1}{2} \times 5 \times 6 \sin C$

M1

$$\sin C = 0.833(3..)$$

AWRT 0.83 or 5 / 6 OE

PI by e.g. seeing 56 or better

A1

(C is obtuse) $C = 123.6^\circ$

AWRT 123.6

A1

3

(b) $\{AB^2 =\} 5^2 + 6^2 - 2 \times 5 \times 6 \cos C$

RHS of cosine rule used

M1

$$= 61 - 60 \times (-0.553...) = 94.1(66...)$$

Correct ft evaluation, to at least 2 sf, of AB^2 or AB using c's value of C.

m1

$$(AB =) 9.7 \text{ (cm to 2sf)}$$

If not 9.7 accept AWRT 9.70 or AWRT 9.71

A1

3

[6]

Q12.

(a) {Arc $\Rightarrow$ } $r\theta = 20 \times 0.8$

$r\theta$ seen in (a) or used for the arc length

M1

$$\dots = 16 \text{ (cm)}$$

A1

2

(b) {Area of sector $\Rightarrow$ } $\frac{1}{2}r^2\theta = \frac{1}{2} \times 20^2 \times 0.8$

$\frac{1}{2}r^2\theta$ *OE seen in (b) or used for the area*

M1

$$\dots = 160 \text{ (cm}^2\text{)}$$

A1

2

(c) {Let $D = \text{angle } ODB$ } $\frac{20}{\sin D} = \frac{15}{\sin 0.8}$

*Sine rule, ACF with $\sin D$ being the only unknown served.
PI by next line*

M1

$$\sin D = \frac{20 \times \sin 0.8}{15} \left\{ = \frac{14.3(471\dots)}{15} \right\}$$

$$\left\{ = \frac{20}{20.9(10\dots)} \right\} = 0.956(474\dots)$$

$$\text{Acute 'D'} = 1.27(467\dots)$$

*Correct rearrangement to ' $\sin D = \dots$ ' or to ' $D = \sin^{-1}(\dots)$ '
OE. PI by at least 3sf correct value 1.27(467...) radians
or $73(.033)^\circ$ for acute angle or PI by at least 3sf
value 1.86(692...) rounded or truncated for D*

m1

$$D = \pi - \text{Acute 'D' in rads}$$

Dep on previous 2 marks being awarded. PI by correct ft evaluation of $\pi - c$'s acute D to at least 3 sf value or seeing 1.86(692...), rounded or truncated, for D

m1

$$\{\text{Angle } ODB\} = 1.87 \{\text{to 3sf}\}$$

Condone > 3sf.

A1

4

[8]

Q13.

$$(8\sec x - 2\sec^2 x = \tan^2 x - 2)$$

$$8\sec x - 2\sec^2 x = \sec^2 x - 1 - 2$$

Using $\tan^2 x = \sec^2 x - 1$ and NOT replacing $\sec^2 x$ with $1 + \tan^2 x$.

M1

$$3\sec^2 x - 8\sec x - 3 (= 0)$$

A1

$$(3\sec x + 1)(\sec x - 3) (= 0)$$

Correct factors or correct use of quadratic equation formula or completing the square for 'their' equation.

$$\sec x - \frac{8}{6} = \pm \sqrt{\frac{64}{36} + 1}$$

m1

$$\text{Or } \sec x = \frac{8 \pm \sqrt{(-8)^2 - 4(3)(-3)}}{2(3)}$$

$$\sec x = 3, -\frac{1}{3} \text{ (or } -0.33)$$

Both correct.

A1

$$\sec x = \frac{1}{\cos x}$$

PI

B1

$$\left(\cos x = \frac{1}{3} \text{ or } 0.33 \right)$$

$$\left(\sec x = -\frac{1}{3} \text{ is impossible} \right)$$

$$x = 1.23, 5.05$$

One correct. Must have earned A1 for correct quadratic, but independent of the second A1.

A1

Both correct and no extras in $0 < x < 2\pi$.

CAO

A1

[7]

Q14.

(a) {Area of sector =} $\frac{1}{2} r^2 \theta = \frac{1}{2} \times 6^2 \times \theta$

$\frac{1}{2} r^2 \theta$ seen in (a) or used for the area

M1

$$21.6 = 18\theta \quad \text{so} \quad \theta = 1.2$$

Must be exact, not rounded to

A1

(b) {Arc =} $r\theta$

$r\theta$ seen in (b) or used for the arc length

M1

$$\dots = 7.2 \text{ {cm}}$$

Ft on $6 \times c$'s value for θ provided $4 < \text{arc} < 10$.

A1F

2

[4]

Q15.

(a) Area = $\frac{1}{2} \times 10 \times AC \sin 150$

$$\frac{1}{2} \times 10 \times AC \sin 150$$

M1

$$40 = 2.5 AC \quad \text{so} \quad AC = 16 \text{ (m)}$$

AG Be convinced

A1

(b) $\{BC^2 =\} 10^2 + 16^2 - 2 \times 10 \times 16 \times \cos 150$

RHS of cosine rule used

M1

$$= 100 + 256 + 277.128 \dots$$

Correct order of evaluation

M1

$$BC = \sqrt{633.128 \dots} = 25.162 \dots = 25.16 \text{ m}$$

AWRT 25.16

A1

(c) $\frac{10}{\sin C} = \frac{BC}{\sin 150} \quad \left(\text{or } \frac{BC}{\sin 150} = \frac{AC}{\sin B} \right)$

A correct equation using sine rule or cosine rule or area formula for either B or C. Subst of BC or AC not required for this M.

M1

$$\sin C = \frac{10 \sin 150}{25.16} \quad (= 0.1987 \dots)$$

Correct rearrangement to either sinC or cosC or sinB or cosB equal to numerical expression ft on c's numerical value for BC. PI by correct C or (by correct B if Mscored)

M1

(or $\sin B = \frac{16 \sin 150}{25.16} \quad (= 0.317 \dots \text{ or } 0.318))$

Smallest angle, (C =) 11.5 ° to 1dp

Accept a value 11.4 to 11.5 inclusive.

A1

Q16.

(a) (i) $\tan \alpha = \frac{4}{3}$

Fraction required

Allow 1.333 (recurring)

B1

1

(ii) 1, 2, $\sqrt{3}$ seen (from Pythagoras)

or

$$4 = 1 + \cot^2 \beta$$

Use $\operatorname{cosec}^2 \beta = 1 + \cot^2 \beta$

M1

$$\tan \beta = -\frac{1}{\sqrt{3}}$$

SC B1 $\frac{1}{\sqrt{3}}$ or $\frac{\sqrt{3}}{3}$

A1

Special case B1 for $\frac{1}{\sqrt{3}}$ or $\frac{\sqrt{3}}{3}$

2

(b) $\tan(\alpha + \beta) = \frac{\frac{4}{3} - \frac{1}{\sqrt{3}}}{1 - \frac{4}{3} \left(-\frac{1}{\sqrt{3}} \right)}$

Use $\tan(\alpha + \beta)$ formula

M1

Remove fractions within fractions

Correct manipulation to form $\frac{a+b\sqrt{3}}{c+d\sqrt{3}}$ $a b c d$ integers

m1

$$= \frac{4\sqrt{3} - 3}{3\sqrt{3} + 4}$$

$m = 4$ $n = 3$
or any multiple

A1

3

Alternative $\tan(\alpha + \beta)$

$$\frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)} = \frac{\frac{4}{5} \times \left(-\frac{\sqrt{3}}{2}\right) + \frac{3}{5} \times \frac{1}{2}}{\frac{3}{5} \times \left(-\frac{\sqrt{3}}{2}\right) - \frac{4}{5} \times \frac{1}{2}}$$

Use formulae for $\sin(\alpha + \beta)$ and $\cos(\alpha + \beta)$

(M1)

Remove fractions within fractions

Correct manipulation to form $\frac{a+b\sqrt{3}}{c+d\sqrt{3}}$ a, b, c, d integers

(m1)

$$= \frac{-4\sqrt{3}+3}{-3\sqrt{3}-4} \left(= \frac{4\sqrt{3}-3}{3\sqrt{3}+4} \right)$$

$m = -4$ $n = -3$
or any multiple

(A1)

(3)

M1 for substituting candidates values for $\tan \alpha$ and $\tan \beta$ into correct formula.
Completely correct or completely correct ft on $\tan \alpha$, $\tan \beta$.

Special case answer is $\frac{12+3\sqrt{3}}{9-4\sqrt{3}}$ or $\times \frac{a}{a}$ where a is integer or $\sqrt{3}$ for M1m1A0.

[6]

Q17.

(a) (Area) = $\frac{1}{2}(26)(31.5)\sin \theta$

$\frac{1}{2}(26)(31.5)\sin(\theta)$ stated or used

M1

$$\frac{1}{2}(26)(31.5) \times \frac{5}{13} = 157.5 \text{ (cm}^2\text{)}$$

OE eg $\frac{315}{2}$

Condone AWRT 157.50

NMS: 157.5 or AWRT 157.50 scores B2

A1

2

(b) $(\cos \theta =) \frac{12}{13}$
 $\frac{12}{13}$ OE exact fraction

B1

1

(c) $\{AC^2 =\}$
 $31.5^2 + 26^2 - 2 \times 31.5 \times 26 \times \cos(\theta) = 992.25 + 676 - 1512$
RHS of cosine rule

M1

$= 1668.25 - 1512 = 156.25$

Correct order of evaluation. Do not award if evaluation leads to or would lead to RHS value being outside interval 120 to 195

m1

$AC = \sqrt{156.25} = 12.5 \text{ (cm)}$

12.5 OE with no sight of premature approximation clearly used

A1

3

Alternative

$\{AC^2 =\} (26 \sin \theta)^2 + (31.5 - 26 \cos \theta)^2$

(M1)

$= 10^2 + 7.5^2$

(m1)

$AC = \sqrt{156.25} = 12.5 \text{ (cm)}$

(A1)

(3)

[6]

Q18.

(a) $\{\text{Arc} =\} r\theta$
 $r\theta$ seen or used for the arc length

M1

$= 18 \times \frac{2\pi}{3} = 12\pi \text{ (m)}$

12π

A1

2

(b) (i) $\alpha = \frac{\pi}{3}$
 $\frac{1}{3} \pi$ OE expression which simplifies to $\frac{1}{3} \pi$

B1

1

(ii) {Area of sector $\Rightarrow \frac{1}{2} r^2 \theta = \frac{1}{2} \times 18^2 \times \frac{2\pi}{3}$
 $\frac{1}{2} r^2 \theta$ seen or used for the sector area

M1

$= 108 \pi (=339.(29..))$

If not exact accept 3sf or better

PI by final correct answer

A1

$\tan \frac{\pi}{3} = \frac{TP}{18}$ {or $\tan \frac{\alpha}{2} = \frac{18}{TP}$ }

{or $PQ = 2 \times 18 \sin \frac{\pi}{3}$ } {or $\frac{1}{2} PQ = 18 \sin \frac{\pi}{3}$ }

EX { or $\cos \frac{\pi}{3} = \frac{18}{OT}$ } { or $\sin \frac{\alpha}{2} = \frac{18}{OT}$ } PRACTICE

OE Correct method (PI) to find either TP or TQ (=TP)

or OT or PQ or $\frac{1}{2} PQ$. If α not $\pi/3$ then ft c's value for

α in (b)(i). If c finds two of TP / TQ, OT and PQ / $\frac{1}{2} PQ$ and gets one correct, one wrong, mark correct one ie M1A1 (M1A0 possible if no correct length)

M1

$TP = 18\sqrt{3} = 31.1769... \text{ exact or } 31.1 \text{ to } 31.2 \text{ incl}$

{or $PQ = 18\sqrt{3} = 31.1769... \text{ exact or } 31.1 \text{ to } 31.2 \text{ incl}$ }

{or $OT = 36$ };

Correct TP or TQ or PQ or $\frac{1}{2} PQ$ or OT either exact value

or in range indicated *PI* by value 561 to 561.3 inclusive for the area of the kite.

A1

$$\frac{1}{2} PQ = 9\sqrt{3} \text{ or } 15.5 \text{ to } 15.6 \text{ incl}$$

$$\text{Area of kite } PTQO = 2 \times \frac{1}{2} \times 18 \times TP$$

$$\{\text{or Area} = \frac{1}{2}(18^2)\sin\frac{2\pi}{3} + \frac{1}{2} TP^2\sin\alpha\}$$

$$\{\text{or area kite} = \frac{1}{2} \times PQ \times \left[18 \div \cos\frac{\pi}{3}\right]\}$$

$$\{\text{or area kite} = \frac{1}{2} \times 2 \times 18\sin\frac{\pi}{3} \times OT\}$$

$$\{= 18^2\sqrt{3}\} \{= 2 \times 162\sqrt{3}\}; \{243\sqrt{3} + 81\sqrt{3}\}$$

*OE valid method to find area of kite, down to a correct expression with no more than 1 unknown length; ft on c's value of α . For method using > one unknown length this *M* is dependent on previous *M* for length*

PI by value $324\sqrt{3}$ or a numerical expression which simplifies to $324\sqrt{3}$; or a value 561 to 561.3 inclusive for the area of the kite. Can also be implied by award of the final A1

M1

Alternative:

$$\text{Area triangle } PTQ = \frac{1}{2} TP^2\sin\alpha \text{ and}$$

$$\text{Area triangle } POQ = \frac{1}{2} 18^2\sin(2\pi/3)$$

*OE Alternative: Award this method mark if **both** area of triangle PTQ ($= 243\sqrt{3}$) and area of triangle POQ ($= 81\sqrt{3}$) are found with or without finding area of kite*

(M1)

Area of shaded region = $561.(18\dots) - 108\pi = 222 \text{ (m}^2\text{) to 3sf A1 6 222.0 inclusive}$

If not 222, condone value from 221.7 to 221.89...

A1

Alternative

$$\begin{aligned}
 \text{Area of shaded region} &= 243\sqrt{3} - (108\pi - 81\sqrt{3}) = 221.89\dots \\
 &= 222 \text{ (m}^2\text{) to 3sf}
 \end{aligned}$$

(A1)

(6)

[9]

Q19.

(a) $\text{Arc} = r\theta$

arc = rθ seen or used. PI by correct θ

M1

$$4 = 5\theta \Rightarrow \theta = \frac{4}{5} = 0.8$$

$$(\theta =) \frac{4}{5} \text{ OE}$$

A1

2

(b) $\text{Area of sector} = \frac{1}{2}r^2\theta$

$$\text{Area} = \frac{1}{2}r^2\theta \text{ seen or used within (b). PI}$$

M1

$$= \frac{1}{2} \times 5^2 \times 0.8 = 10 \text{ (cm}^2\text{)}$$

*Ft on 12.5×c's exact value for θ in part (a)
provided 5 ≤ c's area ≤ 20*

A1F

2

[4]

Q20.

(a) $10^2 = 8^2 + 5^2 - 2 \times 8 \times 5 \cos\theta$

Use of the cosine rule PI by next line

M1

$$\cos\theta = \frac{8^2 + 5^2 - 10^2}{2 \times 8 \times 5} \left(= -\frac{11}{8} = -0.1375 \right)$$

Rearrangement

m1

$$\theta = 97.90(32...) = 97.9^\circ \text{ (to nearest } 0.1^\circ)$$

*CSO (Must see either exact value for $\cos\theta$
or at least 4sf value for either $\cos\theta$ or θ
before the printed answer 97.9°) AG*

A1

3

(b) (i)
$$\text{Area} = \frac{1}{2} \times 8 \times 5 \sin \theta$$

OE

M1

$$= 19.810... = 19.8 \text{ (cm}^2\text{) to 3sf}$$

Condone > 3sf

A1

2

(ii)
$$\text{Area of triangle} = 0.5 \times BC \times AD$$

*Or valid method to find $\sin B$ or $\sin C$
or B or C*

M1

$$AD = [\text{Ans. (b)(i)}] \div [0.5 \times BC]$$

Or $AD = 5 \sin B$; or $AD = 8 \sin C$ OE

m1

$$AD = \frac{19.810..}{5} = 3.962.. = 3.96 \text{ (cm) to 3sf}$$

Condone > 3sf

A1

3

[8]

Q21.

(a)
$$\frac{10}{\sin \theta} = \frac{9}{\sin 54}$$

Sine rule, with $\sin\theta$ being the only unknown

M1

$$\sin \theta = \frac{10 \times \sin 54}{9} \left\{ = \frac{8.09...}{9} \right\} \left\{ = \frac{10}{11.12...} \right\}$$

$$\sin \theta = 0.898(9...) , \quad \theta = 64.01(48...)$$

*Correct rearrangement to 'sin θ = ...' or
to 'θ = sin⁻¹ (...)*

m1

$$\theta = 64^\circ \text{ {to nearest degree}}$$

*AG m1 must have been awarded and must
see at least 3sf value either for sin θ so that
0.898 ≤ sin θ ≤ 0.8993 or for θ so that
64.0 ≤ θ ≤ 64.1 as well as seeing 'θ (OE) = 64'*

A1

3

(b) Angle $C = 180 - (54 + \theta) = 62$ {to 2sf}

$$C = 62. \text{ AWRT } 62.$$

PI if 'C = 180 - (54 + θ)' and accurate later work.

B1

$$\text{Area} = \frac{1}{2} \times 10 \times 9 \sin 62$$

OE Ft c's value for C (C ≠ 54, C ≠ θ)

M1

$$= 39.73.. = 40 \text{ {cm}^2 \text{ to nearest sq cm}}$$

If not 40 condone a value 39.7 to 39.8 inclusive.

A1

3

[6]

Q22.

(a) {Area of sector =} $\frac{1}{2} r^2 \theta = \frac{1}{2} \times 6^2 \times 0.5$

$$\frac{1}{2} r^2 \theta$$

seen within (a) or used for the area

M1

$$= 9 \text{ (cm}^2\text{)}$$

Condone missing/incorrect units

A1

2

- (b) (i) $\{\text{Arc} =\} r\theta = 6 \times 0.5$
 $r\theta$ seen within (b) or used for the arc length

M1

$$= 3 \text{ (cm)}$$

Condone missing/incorrect units

A1

2

- (ii) Perimeter of sector = $6 + 6 + \text{arc length}$
PI by value of 12+c's (b)(i) answer

M1

$$= 15 \text{ (cm)} \quad (= 5 \times 3)$$

Perimeter (of sector) = $5 \times (\text{length of}) \text{ arc}$
Completion, including concluding statement

A1

2

[6]

Q23.

- (a) $h = 0.5$

*$h = 0.5$ stated or used. (PI by x -values
 0, 0.5, 1, 1.5, 2 provided no contradiction)*

B1

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$$f(x) = \sin x$$

$$I \approx h/2\{\dots\}$$

$$\{.\} = f(0) + f(2) + 2[f(0.5) + f(1) + f(1.5)]$$

OE summing of areas of 'trapezia'..

M1

$$\{.\} =$$

$$0 + 0.90929.. + 2[0.4794.. + 0.84147.. + 0.99749..]$$

$$\{.\} = 0.90929.. + 2[2.318..] = \{ 0.90929.. + 4.636..\}$$

*Min. of 2dp values rounded or truncated.
 Can be implied by later correct work
 provided > 1 term or a single term which*

rounds to 1.39

A1

$$(I \approx) 0.25[5.546..] = 1.3865.. = 1.39 \text{ (to 3sf)}$$

CAO Must be 1.39

A1

4

- (b) Stretch(I) in y-direction(II) scale factor 2(III)

Need (I) and either (II) or (III)

M1

All correct. Need (I) and (II) and (III)
[> 1 transformation scores 0/2]

A1

2

(c) $\frac{\sin x}{\cos x} = \frac{1}{2}$; $\tan x = 0.5$

$$\frac{\sin x}{\cos x} = \tan x \quad \text{used to get } \tan x = k$$

or identity $\cos^2 x + \sin^2 x = 1$ used to get either
 $\sin^2 x = p$ or to get $\cos^2 x = q$, (p and q must be
between 0 and 1)

M1

$\tan x = 0.5$

Either $\tan x = \frac{1}{2}$ or $\cos x = \pm \sqrt{\frac{4}{5}}$ ($= \pm 0.894..$)

or $\sin x = \pm \sqrt{\frac{1}{5}}$ ($= \pm 0.447..$)

A1

$x = \alpha$ or $\pi + \alpha$ where $\alpha = \tan^{-1}(k)$

Correct method to find 2nd angle. Any in wrong
ft quadrants then m0. In case of squaring
method candidates must also have rejected
the extra 'quadrants' for the m1. Condone
degrees or mixture

m1

$x = 0.464, 3.61$

Both. Condone > 3sf [0.463(6..), 3.60(5..or 6.)]

Accept pair of truncated values [0.463, 3.60]
 Ignore any answers outside interval 0 to 6.28

A1

4

[10]

Q24.

(a) (i) {Area of sector =} $\frac{1}{2}r^2\theta$
Stated or explicitly used

M1

$$= \frac{1}{2} \times 15^2 \times 1.2 = 135 \text{ (cm}^2\text{)}$$

AG Must see some substitution

A1

2

(ii) {Arc =} $r\theta$
PI

M1

$$\dots = 18 \text{ (cm)}$$

A1

2

(b) $PB = 5 \text{ (cm)}$

*Accept even if only on a diagram or
 within an expression for the perimeter*

B1

$$\{AP^2 =\} 15^2 + 10^2 - 2 \times 15 \times 10 \cos 1.2$$

RHS of cosine rule used

M1

$$= 325 - 300 \cos 1.2 = 216.2926 \dots$$

Correct order of evaluation

m1

$$AP = 14.7(068\dots)$$

PI eg within an expression for perimeter

A1

$$\text{Perimeter} = 5 + 18 + 14.7 \dots = 37.7 \text{ (cm)}$$

3 sf or better

A1

5

[9]

Q25.

(a) $\sin x = \frac{1}{3}$, or sight of ± 0.34 , $\pm 0.11\pi$ or ± 19.47 (or better)

M1

$$x = 0.34, 2.8(0)$$

AWRT

*Penalise if incorrect answers in range;
ignore answers outside range*

A1

2

(b) $\operatorname{cosec}^2 x - 1 = 11 - \operatorname{cosec} x$

Correct use of $\cot^2 x = \operatorname{cosec}^2 x - 1$

M1

$$\operatorname{cosec}^2 x + \operatorname{cosec} x - 12 (= 0)$$

A1

$$(\operatorname{cosec} x + 4)(\operatorname{cosec} x - 3) (= 0)$$

Attempt at Factors

*Gives cosec x or - 12 when expanded
Formula one error condoned*

m1

$$\left. \begin{array}{l} \operatorname{cosec} x = -4, 3 \\ \sin x = -\frac{1}{4}, \frac{1}{3} \end{array} \right\}$$

Either Line

A1

$$\sin x = -\frac{1}{4}$$

$$\Rightarrow x = 3.39, 6.03$$

AWRT

3 correct or their two answers from (a)

and 3.39, 6.03

B1F

0.34, 2.8(0)

AWRT

*4 correct and no extras in range
ignore answers outside range*

SC 19.47, 160.53, 194.48, 345.52 B1

B1

6

Alternative

$$\frac{\cos^2 x}{\sin^2 x} = 11 - \frac{1}{\sin x}$$

$$\cos^2 x = 11 \sin^2 x - \sin x$$

*Correct use of trig ratios and multiplying
by $2 \sin^2 x$*

(M1)

$$1 - \sin^2 x = 11 \sin^2 x - \sin x$$

$$0 = 12 \sin^2 x - \sin x - 1$$

(A1)

$$0 = (4 \sin x + 1)(3 \sin x - 1)$$

Attempt at factors as above

(m1)

$$\sin x = -\frac{1}{4}, \frac{1}{3}$$

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As above

(B1F)(B1)

[8]

Q26.

(a) $R = \sqrt{10}$

Accept $R = 3.16$ or better

B1

$$\tan \alpha = 3$$

OE

M1

$\alpha = 1.249$ ignore extra out of range
AWRT 1.25 SC $\alpha = 0.322$ B1
radians only

A1

3

(b) (i) minimum value = $-\sqrt{10}$
F on R

B1F

1

(ii) $\cos(x - \alpha) = -1$

M1

$x = 4.391$
AWRT 4.39
51.56° or ... 57° or better

A1F

2

(c) $\cos(x - \alpha) = \frac{2}{\sqrt{10}}$

M1

$x - \alpha = \pm 0.886$ 5.397
 ignore extra out of range
Two values, accept 2dp and condone 5.4
condone use of degrees

A1

$x = 0.36296..$ 2.13512..
F on $x - \alpha$, either value. AWRT

A1F

$x = 0.363$ 2.135
CSO *3dp or better*

A1

Alternative

$$10 \sin^2 x - 12 \sin x + 3 = 0$$

*Or equivalent quadratic using $\cos x$
 (ie $\sin^2 x + \cos^2 x = 1$ used)*

M1

$\sin x =$ two numerical answers

Or equivalent using $\cos x$

A1F

$$-1 \leq \text{ans} \leq 1$$

$x =$ one correct answer

A1F

$$x = 0.363 \quad 2.135$$

CSO

3 dp or better

A1

4

[10]

Q27.

(a) {Area of sector =} $\frac{1}{2} r^2 \theta$
 $\frac{1}{2} r^2 \theta$ *seen or used for the area*

M1

$$= \frac{1}{2} \times 8^2 \times 1.4 = 44.8 \text{ {m}^2 \}$$

Must be exact, not rounded to

A1

2

(b) (i) {Arc =} $r\theta$

$r\theta$ seen or used for the arc length

M1

$$\dots = 11.2$$

PI Condone AWRT 11.2

A1

$$\text{Perimeter of sector} = 16 + 11.2 = 27.2 \text{ {m} \}$$

Ft on c's evaluation of 8×1.4

A1F

3

(ii) $27.2 = 2\pi x$

[c's numerical answer for (b)(i)] = $2\pi x$

M1

$$x = \frac{27.2}{2\pi} = 4.329... = 4.33 \text{ to 3sf}$$

Condone > 3 sf

A1

2

[7]

Q28.

(a) $\frac{6}{\sin \theta} = \frac{15}{\sin 150}$

Sine rule OE PI

M1

$$\sin \theta = \frac{6 \times \sin 150}{15} \quad \{= 0.2\}$$

Rearrangement

m1

$$\theta = 11.53(6..) = 11.5^\circ \text{ {to nearest } 0.1^\circ \}$$

AG Must see at least 4 sf value or an exact value for $\sin \theta$ (0.2, 3/15, OE) before seeing the printed value 11.5

A1

3

(b) Angle $B = 180 - (150 + \theta) = 18.5 \text{ {to 3 sf}}$

Award for B = any value between 18 and 19 inclusive [18.463041....]

B1

$$\text{Area} = \frac{1}{2} \times 6 \times 15 \sin B$$

M1

$$= 14.3 \text{ cm}^2 \text{ to 3sf}$$

Accept a value 14.2 to 14.3 inclusive
Note: For methods involving AC, for the M1 need both a correct method to find AC and a correct area formula

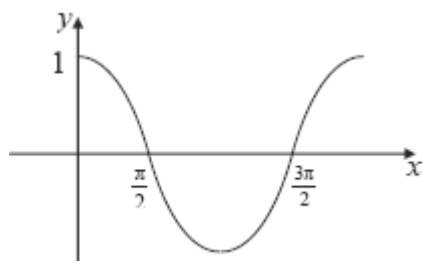
A1

3

[6]

Q29.

(a)



Correct shape meeting positive y-axis
and only one oscillation within interval
0 to 2π

B1

The three correct intercepts stated;
Accept 1.57 for π/2 and 4.71 for 3π/2
but must be evidence of radian vals.
not just degrees

B1

Ignore any parts of the graph clearly
indicated as outside the given interval

2

(b) (i) $1 - \cos^2 \theta = \cos \theta(2 - \cos \theta)$
 $\cos^2 \theta + \sin^2 \theta = 1$ used

M1

$$1 = 2 \cos \theta \Rightarrow \cos \theta = \frac{1}{2}$$

CSO AG Completion

A1

2

(ii) $\sin^2 2x = \cos 2x(2 - \cos 2x) \Rightarrow \cos 2x = \frac{1}{2}$
Uses (b)(i)

M1

$\{2x =\} \cos^{-1}\left(\frac{1}{2}\right) = 1.04(7..)$

PI Accept $1.05, \frac{\pi}{3}$; Condone 60°

m1

$x = 0.524, 2.62$

$x = 0.523(59..), 2.61(7..)$

Condone > 3sf; Condone $x = 0.525, 2.62$

Accept truncated '3sf' vals $x = 0.523, 2.61$

Deduct 1 mark for each extra (> 2 solns) in the given interval from A marks to a min of A0. Ignore any solns outside the given interval 0 to π .

Accept, as equivalent, the exact answers

$x = \frac{\pi}{6}$ and $x = \frac{5\pi}{6}$ when seen and apply

ISW if 'errors' converting these to decimals.

If not A2 then A1 if

- one soln correct.
- $30^\circ, 150^\circ$ ie solns left in degrees
- AWR 0.52, 2.6 ie correct vals to only 2sf.

Must see an indication that (b)(i) has been used otherwise 0/4 so just stating the two correct answers with nothing else scores 0/4.

A2,1,0

4

[8]

Q30.

(a) {Area of sector =} $\frac{1}{2}r^2\theta$
 $\frac{1}{2}r^2\theta$ stated or used for area of sector. PI

M1

$$= \frac{1}{2} \times 10^2 \times 0.8 = 40 \text{ {cm}^2}$$

A1

2

- (b) (i) {Arc =} $r\theta$
 $r\theta$ stated or used for arc length. PI

M1

$$\dots = 8$$

PI

A1

$$\text{Perimeter} = 20 + r\theta = 28 \text{ (cm)}$$

ft on $20 + r \times \theta$

A1Ft

3

- (ii) Area of square = $\left[\frac{\text{c's answer for (b)(i)}}{4} \right]^2$
PI

M1

$$= 49 \text{ {cm}^2}$$

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A1cao

2

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[7]

Q31.

- (a) {Area =} $\frac{1}{2} \times 7.4 \times 5.26 \times \sin 63^\circ$

M1

$$= 17.3(407\dots) \text{ {m}^2}$$

Accept any value from 17.3 to 17.341

A1

2

- (b) { BC^2 =} $5.26^2 + 7.4^2 - 2 \times 5.26 \times 7.4 \cos 63$
RHS of cosine rule used

M1

$$\dots = 27.66(76) + 54.76 - 35.34(22\dots)$$

Correct order of evaluation

m1

$$\Rightarrow BC = \sqrt{47.08(5\dots)} = 6.861(8\dots)$$

$$BC = 6.86 \text{ {m} to 3sf}$$

AG. Cand. must show a 4th sf in either

*$\sqrt{47.08(5\dots)}$ or 6.861(8) before giving the
printed answer 6.86*

A1

3

(c) $\frac{\sin B}{5.26} = \frac{\sin 63}{BC}$

*Sine rule involving 'sin B' [If valid cosine
rule used to find cos B, no marks awarded
until stage of converting to sin B]*

M1

$$\sin B = 0.68 \text{ to 2sf}$$

*If not 0.68, accept AWRT any value from
0.682 to 0.684 inclusive*

A1

ALTn

$$\frac{1}{2} \times 7.4 \times (6.86\dots) \times \sin B = c' \text{ s ans (a)}$$

(6.86 ..) could be c' s ans (b)

(M1)

$$\sin B = 0.68 \text{ to 2sf}$$

*If not 0.68, accept AWRT any value from
0.682 to 0.684 inclusive*

(A1)

2

[7]

Q32.

(a) $\{x = \} \sin^{-1}(0.8) = 0.927(29\dots) \{= \beta\}$

$$\sin^{-1}(0.8) \text{ PI}$$

M1

$$\{x = \} \pi - \beta$$

m1

$$x = 0.927(29...), \quad 2.21(42...)$$

Both

Ignore values outside interval $0 - 2\pi$ but

A0 if 'extra' values inside the given interval

A1

3

(b) (i) $\left(\frac{3\pi}{2}, -1\right)$

B1 if one coordinate correct or $\left(-1, \frac{3\pi}{2}\right)$

B2,1

2

(ii) $\pi - \alpha$

B1

1

(iii) $RS = (2\pi - \alpha) - (\pi + \alpha)$

OE eg $RS = PQ = (\pi - \alpha) - \alpha$

M1

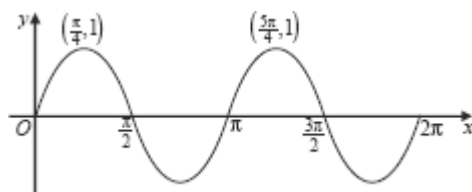
$$= \pi - 2\alpha$$

Must be simplified

A1

2

(c)



Sine curve with positive gradient at O
with at least 3 stationary points between 0
and 2π

B1

Correct shaped curve with 2 max and 2 min between 0 and 2π

B1

All 5 correct points of intersection with

x -axis with $\frac{\pi}{2}$, π and $\frac{3\pi}{2}$ clearly shown

B1

Maximum points $\left(\frac{\pi}{4}, 1\right)$ and $\left(\frac{5\pi}{4}, 1\right)$
 stated or clearly shown on the sketch

B1 for either:

1 as the y -coordinate of max pt(s) or:
 two max pts between 0 and 2π with
 correct x -coordinates

B2,1

5

[13]

Q33.

(a) $R = \sqrt{10}$

Accept $R = 3.16$ or better.

B1

$\tan \alpha = 3$

Oe (Can be implied by 71.57° seen)

M1

$\alpha = 1.25$

A0 if extra answers within given range

SC 1 $\tan \alpha = \frac{1}{3}$ $\alpha = 0.32$

A1

3

(b) (i) min value = $-\sqrt{10}$ (or $\geq \sqrt{-10}$)
 ft or R

B1F

1

(ii) $\sin(x - \alpha) = -1$

$$x = 5.96$$

$$\text{or } \sin^{-1} \frac{3\pi}{2}$$

M1

$$Ft \text{ on their } \alpha \text{ (to 2 dp)} + \frac{3\pi}{2}$$

A1F

2

[6]

Q34.

(a) $5^2 = 7^2 + 8^2 - 2 \times 7 \times 8 \cos \theta$

*Use of the cosine rule – must be correct
(PI by the correct line below)*

M1

$$\cos \theta = \frac{7^2 + 8^2 - 5^2}{2 \times 7 \times 8} (= \frac{88}{112} = 0.7857 \dots)$$

Rearrangement

m1

$$\theta = 38.21\dots = 38.2^\circ \text{ (to nearest } 0.1^\circ)$$

*CSO (Must see either exact value for
 $\cos \theta$ or at least 4sf value for either $\cos \theta$
or θ before the printed answer 38.2°) AG*

A1

3

(b) Area = $\frac{1}{2} \times 7 \times 8 \sin \theta$

$$\text{OE eg Area} = \sqrt{10(10-5)(10-8)(10-7)}$$

$$(\text{= } \sqrt{300})$$

M1

$$= 17.3 \text{ \{cm}^2\} \text{ to 3 sf}$$

Condone 17.31 to 17.33 inclusive

A1

2

[5]

Q35.

$$\{\text{Area of sector} = \frac{1}{2} r^2 \theta$$

$$\frac{1}{2} r^2 \theta$$

seen or used for the area; PI

M1

$$r^2 = \frac{33.75}{\frac{1}{2} \theta} (= 56.25)$$

Correct rearrangement to $r^2 = \dots$ or $\dots r = \dots$

m1

$$r = 7.5$$

PI eg by a correct arc length

A1

$$\{\text{Arc} = \} r\theta$$

$r\theta$ seen or used for the arc length

M1

$$\dots = 9$$

*ft on $1.2 \times \text{cand's } r$ provided the **two** M's scored; if not explicit, PI by ft on $3.2 \times \text{cand's } r$ for perimeter*

A1F

$$\{\text{Perimeter} = \} 24 \{\text{cm}\}$$

CAO

A1

[6]